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Module Objectives
Prime Factorization
RSA Encryption and Decryption
RSA Key Setup
Security of RSA

Video: Security of RSA
5 min

Reading: Lecture Slides for RSA Algorithm
15 min

Quiz: RSA Algorithm
Submitted

Reading: Paper by Rivest, Shamir, and Adleman
50 min

RSA Algorithm

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Review Learning Objectives

1. Using the same notations as in the lecture, p and q are the two prime factors of n, and e and d are the public key and the private key, respectively. Which of the followings are to be secret against the attacker, assuming that the Prime Factorization problem remains difficult to solve?

4 / 4 points

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Due Apr 21, 11:59 PM IST **Attempts** 3 every 8 hours

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- [x] The Euler totient function of n, phi(n)
  - **Correct** See RSA Key Setup lecture.
- [x] q
  - **Correct** See RSA Key Setup lecture.
- [x] d
  - **Correct** See RSA Key Setup lecture.
- [x] p
  - **Correct** See RSA Key Setup lecture.
- [ ] The Euler totient function of e, phi(e)
- [ ] The Euler totient function of p, phi(p)
- [ ] e
- [ ] n

2. Using the same notations as in the lecture, p and q are the two prime factors of n, and e and d are the public key and the private key, respectively.

Which of the following statements about the RSA cipher are true? Select all that applies.

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- [x] After choosing e, the Extended Euclidean algorithm can be used to derive d.
  - **Correct**
- [ ] e and d are independent from p and q.
- [ ] RSA encryption and decryption computations (from plaintext to ciphertext and from ciphertext to plaintext) use all of p, q, e, and d.
- [x] After choosing d, the Extended Euclidean algorithm can be used to derive e.
  - **Correct**
- [x] For the public-private keys of RSA, e and d, given any plaintext m, m raised to the power of e.d (m^{e.d}) is equal to m.
  - **Correct**

3. Using the same notations as in the lecture, e.g., p and q are the two primes and e and d are the public key and the private key, respectively, p=5, q=11. Alice chooses a private key d and derives the public key e. Which of the followings can work for the values of d?

4 / 4 points

- [x] 17
  - **Correct**
- [ ] 8
- [ ] 5
- [x] 9
  - **Correct**
- [ ] 2
- [x] 21
  - **Correct**

4. Using the same notations as in the lecture, e.g., p and q are the two primes and e and d are the public key and the private key, respectively, p=3, q=11, and e=7.

The attacker performs a chosen-ciphertext attack (CCA). It has a known ciphertext 14 and wants to retrieve the corresponding plaintext. It computes c' = c * r^e mod n and chooses the ciphertext 14 * 2187 mod 33 = 27 and retrieves the corresponding plaintext, 15. What is the plaintext corresponding to the ciphertext 14?

4 / 4 points

- (•) 5
- ( ) 12
- ( ) 37
- ( ) 23
- ( ) 3

Correct